

To intervene or not to intervene

Anonymous ACL submission

Abstract

This document is a supplement to the general instructions for *ACL authors. It contains instructions for using the L^AT_EX style files for ACL conferences. The document itself conforms to its own specifications, and is therefore an example of what your manuscript should look like. These instructions should be used both for papers submitted for review and for final versions of accepted papers.

1 Introduction

Dimitris, Katerina

2 Related Work

2.1 Human facilitators

Katerina, Dimitris Add research on how humans typically behave when facilitating.

2.2 Facilitation detection

Dimitris Add work that already tried to answer this question (Fora, WHoW, the second whow aussie paper).

3 Dataset

3.1 Description

[Korre et al. \(2025\)](#) provide a list of datasets relating to discussion facilitation. We combine all of these datasets, preprocess them (A.1) and standardize their data according to a simple, unified schema (A.2). We thus create a multi-domain dataset focused around facilitation, called the Prosocial and Effective Facilitation in Konversations (PEFK). Unlike its predecessors, PEFK is both rich in size and comment-level information, allowing us to run thorough analysis between similar datasets.

3.2 Text-based vs. transcribed datasets

Dimitris, Katerina Our datasets are fundamentally different depending on whether the data are tran-

scripts or actual text data. Dimitris will add technical examples (i.e., how the transcriptions have a different format, e.g., comments that cut each other off). Katerina will add social science research on how oral and written speech differs. Thus, we run separate experiments for each of the two dataset types.

4 How to decide when to intervene?

4.1 Do not decide

Dimitris. What most platforms do is to use flagging and reports to decide when to intervene. Mention Grok as a facilitative agent that works by invocation only. Briefly talk about advantages and disadvantages of current approach.

4.2 Use humans

4.2.1 Merits and disadvantages

Katerina. Humans may be the “best” classifier since they understand context and social behavior best, but they are not scalable (cite our survey). Dimitris can add cost calculations for a hypothetical scenario to drive point home.

4.2.2 Experimental setup

Katerina. Annotation study. We leave experimental details for the appendix, but mention expert annotators, basic procedure, data and setup here. Also talk about positive vs negative reinforcement.

4.2.3 Results

Katerina. Inter-annotator agreement and discussion.

4.3 Use LLMs

4.3.1 Can we detect facilitation interventions?

Dimitris. We use LLMs and transformer-based models to decide whether the person talking is a facilitator.

Name	#comments	Domain	Format
WikiDisputes (2021)	96,320	Forum	Text
WikiTactics (2022)	3,850	Forum	Text
WikiConv (2018)	17,806,373	Forum	Text
Conversations Gone Awry / CMV II (2019)	40,607	Forum	Text
CeRI (2012)	3,700	Forum	Text
User Moderation (UMOD) (2024)	2,000	Forum	Text
Intelligence Squared 2 (IQ2) (2016)	34,245	Debate	Oral
Why How Who (WHoW) (2025)	25,542	Radio / TV	Oral
Fora (2024)	39,438	Deliberation	Oral

Table 1: The datasets used in PEFK. We include a large variety of domains, and two different sources; text-only datasets, and transcripts from oral discussions.

4.3.2 Can LLMs decide when to intervene?

Dimitris, Katerina. No. We show this with two experiments. One is agreement between humans and LLMs on the annotation task (strong baseline). The other is by predicting when a facilitator would speak (weak baseline).

5 Conclusion

Humans are good but expensive. Our research is good. We wasted a good amount of money on this, please accept this.

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A Dataset details

A.1 Preprocessing

General

- We exclude comments with no text.
- We exclude discussions with less than two distinct participants.
- We exclude discussions which are common between wiktactics and wikiconv as well as wikidisputes and wikiconv.
 - There may be duplicate discussions between wikidisputes and wiktactics, but we allow them since they feature complementary information.

Wikiconv

- The Wikiconv corpus does not contain information about which user is a moderator/facilitator. Therefore, all comments relating to Wikiconv are tagged as non-moderators.
- Additionally, we follow the instructions of the original researchers (Hua et al., 2018), and select only discussions which have at least two comments by different users.
 - Wikipedia (thankfully) does not track users who log in with only an IP address (in the original dataset, their user_id is always set to 0 and their username is of the form 211.111.111.xxx. We consider each such username to be a separate user.
 - Due to the size of the dataset, we have to partially load it during preprocessing. Thus, there is a small chance every 100,000 records that a discussion is marked as a false negative and a part of it gets discarded.
 - We only include English comments in the final dataset. We use a small, efficient library (py3langid) for language recognition, due to the large size of Wikiconv. Non-English comments are discarded before selecting valid discussions (see point above).

Wiktactics

- We infer facilitative actions by whether the comment belongs in any of the following categories:
 - Asking questions
 - Coordinating edits
 - Providing clarification

– Suggesting a compromise

– Contextualisation

- The above tactics are a subset of the Coordinative labels used in the WikiTactics paper. They were selected because they are not used necessarily on 1-1 discussions; they could reasonably be applied by third-party participants. Contrast them with other Coordinative labels such as “Conceding/recanting” and “I don’t know”.

Wikidisputes

- Since only 0.03% of the comments in the dataset are made by moderators, we mark the dataset as not supporting moderation.

UMOD

- Facilitative actions are marked as a gradient from 0 (no facilitation) to 1 (full facilitation). We adopt a threshold of 0.75 to consider an action as facilitative, with more than 50% annotator agreement (measured as entropy in the original dataset).

CMV-AWRY2

- We mark a discussion as escalated when the derailment value (from the official dataset) is in the 60th upper percentile.
- We remove deleted (“[deleted]”) comments.

A.2 Dataset Schema

Table 2 details the information available for each comment in the dataset. Some information, such as whether the person talking is a moderator / facilitator, is available only on some sub-datasets. Also note that the is_moderator column may hold different definitions of what we consider as moderator / facilitator. UMOD for instance considers facilitative comments made by users, while others such as Fora consider a comment facilitative if it was made by a dedicated facilitator.

Name	Type	Description
conv_id	string	The discussion’s ID. Comments under the same discussion refer to the same discussion ID.
message_id	string	The message’s (comment’s) unique ID.
reply_to	string	The ID of the comment which the current comment responds to. nan if the comment does not respond to another comment (e.g., it’s the Original Post (OP)).
user	string	Username or hash of the user that posted the comment
is_moderator	bool	Whether the user is a moderator/facilitator.
moderation_supported	bool	True if the moderation labels are directly computed from the original dataset
escalated	bool	A discussion-level measure denoting discussions which have been derailed
escalation_supported	bool	True if the escalation labels are directly computed from the original dataset
text	string	The contents of the comment
dataset	string	The dataset from which this comment originated from
notes	JSON	A dictionary holding notable dataset-specific information
toxicity	float	The “toxicity” score given to the comment by the Perspective API
severe_toxicity	float	The “severe toxicity” score given to the comment by the Perspective API

Table 2: The full set of columns used for each comment in the PEFK dataset.